Advice After Introductory Physics

If you've self-studied calculus-based physics, here are some resources for going further. For advice on how to start out learning physics or doing physics competitions, see my introductory advice file.

Deepening Core Knowledge

You're done with the introductory material after finishing a book like Halliday, Resnick, and Krane. If you enjoyed what you learned, there are many ways to deepen your knowledge.

- For a deeper understanding of mechanics and special relativity, try *An Introduction to Mechanics* by Kleppner and Kolenkow or *Introduction to Classical Mechanics* by Morin, used for honors introductory physics at MIT and Harvard, respectively. Though they cover similar ground, Morin takes a more synthetic approach, focusing on frictionless planes and massless strings, while Kleppner and Kolenkow discuss cars and rockets.[1]
- For a deeper understanding of electromagnetism, try *Electricity and Magnetism* (3rd edition) by Purcell and Morin, used for honors introductory physics at many top colleges. It has many insightful discussions and problems, but it's best known for a beautiful chapter that uses relativity to introduce magnetism. For a short and sweet introduction to vector calculus that complements the book, try Schey's *Div, Grad, Curl, and All That*, which has good pictures; for a more thorough introduction, see MIT OCW's 18.02.
- The waves and thermodynamics chapters in Halliday, Resnick, and Krane are a good start, but for more content aimed explicitly for Olympiad preparation, try the first volume of *Competitive Physics* by Wang and Ricardo for waves, and the second volume for thermodynamics.
- It's rewarding to get a broad overview of physics as an intellectual tradition. Try the *Feynman Lectures on Physics* for fascinating discussion from a master of the subject.

Core Undergraduate Content

Moving beyond the Olympiad syllabus, you can start exploring the content of a physics degree.

- The standard next thing to do is learn about waves and differential equations. This corresponds to MIT OCW's 8.03, with further background on differential equations given in 18.03. The old 8.03 lectures have additional neat demonstrations. For a textbook reference, *Vibrations and Waves* by French is short and clear, covering the essentials of the subject, while *Waves* by Crawford is more thorough, with many real-world applications and home experiments.
- For intermediate electromagnetism, the best book by far is *Introduction to Electrodynamics* by Griffiths. It's very clear, and its problems get surprisingly deep, due to Griffiths' experience writing many papers on the subtleties of classical electromagnetism. (The 4th and 5th editions are nearly identical, though the new formatting of the 5th is a bit ugly.) This level of electromagnetism requires deeper familiarity with vector calculus, which you can pick up from the

[1] Personally, I prefer Morin's discussion of relativistic kinematics (where his discussion is extensive, and the synthetic approach is very clean) but prefer Kleppner and Kolenkow for everything else. Morin's problems are often too similar, or introduce unnecessary complications; for instance, there are far too many problems involving big stacks of pulleys. Math students tend to like Morin problems because they avoid conceptually tricky points where familiar approaches break down, but identifying and understanding such issues is the very heart of physics.

first few chapters. For a dedicated introduction to the math, see David Tong's notes on Vector Calculus for a clean approach that gets to tensors[2] and introduces index notation, a powerful tool for deriving more complicated vector calculus identities.

- Intermediate mechanics, involving the Lagrangian and Hamiltonian, is not useful for the Olympiad, but extremely important for physics in general. For a clear and gentle introduction, see *Classical Mechanics* by Taylor. For a nice supplement, see David Tong's notes on Dynamics.

- These books are clear, but lack truly hard problems. They can give you the mistaken impression that more advanced physics just consists of routinely applying an ever-longer list of standard formulas. For instance, most undergraduate mechanics books cover the formalism of Hamiltonian mechanics for 100 pages, but then don't use it to solve anything beyond the harmonic oscillator. Of course, the 19th century giants loved tough problems, and they invented these theoretical tools precisely to solve them. To see this side of physics, try flipping through *Analytical Dynamics of Particles and Rigid Bodies* by Whittaker, and *Static and Dynamic Electricity* by Smythe. Both include many problems from the legendary Cambridge Tripos exam, which was an order of magnitude harder than any graduate exam today. The latter was designed, according to Smythe, to "weed out weaklings" in the Caltech graduate program. Unfortunately, these problems assume familiarity with topics no longer taught, like elliptic functions and parabolic coordinates. If you find Smythe too hard, Jackson's book is relatively easy in comparison.

- For intermediate thermodynamics, try either *Concepts in Thermal Physics* by Blundell and Blundell, or *Thermal Physics* by Schroeder. Blundell and Blundell is more comprehensive, with many interesting sidenotes and applications; Schroeder is short, clear, and crisp.

- Don't use "study guide" books marketed as "student friendly", "for students", or even "for dummies". They all have glowing reviews[3], but they don't explain how anything works; they just tell you recipes to solve the simplest exam problems. In a course, they can help if you're short on time and only interested in grades, but the superficial understanding you'll get from them will collapse by the first lecture of the next course. If you're self-studying, there's no reason to ever use these books, because you have the time to do it right.

The above books are just appetizers for the main course of a physics major: quantum mechanics.

- You don't need any book titled "Modern Physics." These books have the same problem as algebra-based physics textbooks: they try to explain too much with too little. Results that should be derived with two lines of math end up "motivated" with hundreds of vague words. The problems almost all reduce to plugging numbers into a given formula. You don't need this kind of book for the Olympiad, nor to start learning quantum mechanics properly.

[2] By the way, tensors are another one of those topics where you need an "official" source, like a book chapter or a polished set of lecture notes. When I google "introduction to tensors" in 2026, most of the first ten results are terrible for beginners. I recognize a few that once got me confused for days, because they were full of typos, and cross-checking them made things worse because they used incompatible conventions. Others are correct, but way too sophisticated to serve as a real introduction. It doesn't help that mathematicians, computer scientists, engineers, and physicists all mean different things when they say "tensor"! Don't try to learn this kind of thing by opening twenty tabs. Just use one good source and stick with it. Exploring will deepen your knowledge only once you have the foundation set.

[3] These reviews make it hard to differentiate quality. For example, A Student's Guide to Maxwell's Equations is okay, but A Student's Guide to Lagrangians and Hamiltonians is terrible: its derivations are vague or wrong, and half the problems aren't properly formulated. I've seen many StackExchange questions from students misled by the latter book. But both have a hundred five-star ratings online.

- There are two entry points to quantum mechanics. The traditional way is to start with using Schrodinger's equations to solve for wavefunctions; this lets you seamlessly transition into it after finishing a course on waves, and supplies a lot of visual intuition. A newer approach is to start by applying the fundamental postulates to low-dimensional systems, like the qubits of quantum computing; this makes the subject accessible to people with no physics background at all, but it is somewhat abstract and requires familiarity with linear algebra. Of course, if you want to really know quantum mechanics, you'll have to understand both perspectives.

- To start the first way, read *Introduction to Quantum Mechanics* by Griffiths, which comes with clear explanations and great problems. For video lectures at this level, try the MIT OCW 8.04 lectures by Barton Zwiebach (clearer) and Allan Adams (more energetic).

- To start the second way, read *Introduction to Linear Algebra* by Strang, used for MIT OCW's 18.06. (This is an excellent thing to do in general, even if you don't stay in physics, because linear algebra is arguably even more useful than calculus. You'll still need it if you go into computer science or finance.) Then read *A Modern Approach to Quantum Mechanics* by Townsend.

- More advanced quantum mechanics courses will freely use both perspectives. For material at this level, see *Principles of Quantum Mechanics* by Shankar, or the MIT OCW 8.05 lectures.

Graduate Topics

I don't recommend jumping to flashy topics like general relativity, quantum field theory, or string theory too early; it is much better to learn graduate-level electromagnetism and quantum mechanics thoroughly first. (See here for a roadmap to that point.) If you're clever, you might be able to brute force your way through some problems in books about the flashy topics, by mechanically applying rules, but you won't have the conceptual foundation to understand why the rules are right. Still, many young students are excited to start these topics, so here are some recommendations.

- Don't take popular science books, such as *The Elegant Universe* or *A Brief History of Time*, too seriously. These books are built on fragile analogies that aren't used by practitioners, and don't give you the power to work out anything new for yourself.[4] If doing problems is practicing an instrument, then a textbook is an inspiring performance, while a popsci book is just pictures of a performance. It's a shadow of the real thing, missing what makes it beautiful.

- For general relativity, the two gentlest books are *Gravity* by Hartle and *A First Course in General Relativity* by Schutz. While both cover similar ground, Schutz puts all the mathematical background up front (including a great introduction to four-vectors and tensors), while Hartle starts with physical results, having you take some of the math on faith until it's filled in later. Both are good, so just pick whichever style you prefer.

- Leonard Susskind's Theoretical Minimum lectures cover graduate-level topics using just calculus. They don't replace a real introduction, but they'll give you the flavor of the underlying logic.

- For particle physics, David Tong's Particle Physics lecture notes cover the basics of the Standard Model and beyond, with many references to the history and deeper theory, and only high school

[4] Even for the things that don't require math, it's easy to misunderstand without knowing what the words actually mean, because English is ambiguous. For example, licorice is better than nothing, and nothing is better than ice cream. So is licorice better than ice cream?

math. Griffiths' *Introduction to Elementary Particles* clearly explains the basics of particle physics and the structure of the Standard Model, along with how to do some toy calculations in quantum field theory; it requires undergraduate quantum mechanics and special relativity.

- For string theory, Cumrun Vafa's *Puzzles to Unravel the Universe* is like a popsci book, but with points explained with neat mathematical puzzles. It's meant for freshmen at Harvard, and assumes only basic algebra. Barton Zwiebach's *A First Course in String Theory* is the string theory textbook with the least prerequisites, and it actually derives many of the results it uses; it assumes undergraduate quantum mechanics and special relativity.

Learning Traps

A lot of online advice about learning physics is quite poor, so here are some traps to avoid.

- Taking an unnecessarily hard route. Difficulty doesn't produce understanding, thinking does. If you read a book upside down for 100 hours, you'll probably have a better understanding than somebody who skimmed through it in 5 hours, but you'll have a worse understanding than somebody who read it normally for 20 hours.[5] Similarly, in the time it takes to work through a famous advanced book without the prerequisites, you could work through both the prerequisites *and* the advanced book, if you do it in the natural order. Life is too short for artificial difficulty.

- Privileging the oldest sources, such as Newton's *Principia Mathematica*, Dirac's quantum mechanics book, or Einstein's papers in the original German. This might be a reasonable heuristic for philosophers, since human nature hasn't changed, but in physics, old sources are a bad place to start. They contain tons of discussion which has turned out to be misguided or irrelevant, and their notation is usually clunky and obsolete. Original sources are works of genius, but modern textbooks are the result of hundreds of geniuses refining those arguments over decades or centuries, and they're designed to be learned from.

- Trying to find the single most comprehensive book possible. For example, many people buy *The Road to Reality* because it promises to cover all of physics in a single book. But I haven't encountered anybody who actually learned something new from it; it's a brilliant and masterful review, but it only helps if you already know the material. As another example, *Gravitation* by Misner, Thorne, and Wheeler is an impressively *large* general relativity textbook. It weighs six pounds, runs to 1300 pages, and uses extra large paper with large font and large margins, so that you can't even open it without the spine falling apart. These features tend to attract ambitious students. But the book is from 1973, which puts it before modern observational cosmology, precision solar system and terrestrial tests, and the experimental discoveries of black holes and gravitational waves. It also doesn't cover that much material: if you convert it to normal font, and strip out the hundreds of pages of vague, equation-free speculations (which have not held up over time), it is no bigger than any modern textbook[6]. Older professors treat this book with respect because it's from the "golden age" of relativity, which led to the advances of the past 50 years. But learning from it today is like learning electromagnetism from Maxwell's *Treatise*. There's nothing great about huge books; three good books do more than one bloated one.

[5] There's also a selection effect here: only the strongest students will have the energy to work through a whole book upside-down. But that doesn't *make* them strong; they'd be even stronger if they worked through it normally.

[6] When it was published, Chandrasekhar wrote: "There is needless repetition (indeed, almost everything is stated at least three times) and the style fluctuates from precise mathematical rigor to evangelical rhetoric (which often obscures and confuses the issues)."

- Following the longest list of books. For example, Gerard 't Hooft's How to become a GOOD Theoretical Physicist is a massive list of resources that were available on the internet in the 1990s. People like it because it looks long and comprehensive, but half the links are broken, and the rest point to abandoned, unfinished drafts of lecture notes from random universities. I don't know anyone who has productively used it. (If you need to find a very specific resource, not covered above, you can try using this list on Physics StackExchange, or asking AI.)

- Jumping on the newest resources online. Physics changes over decades, but not from year to year. A well-checked, student-tested 10-year-old book will almost always beat something written on the spot, and the best physics blog posts (while fun) contain less insight than a single chapter in an average book. Nowadays, there are thousands of people advertising learning apps they made in a day using AI, with fancy graphics but little actual content. As of 2026, AI can do a decent job of explaining individual pieces of standard course content, but to make a good textbook one has to combine a thousand such pieces together, organized in a logical and consistent way, along with a rich set of practice problems. I think this is possible in principle, but I haven't seen anybody do it yet, and it costs much more than just buying a good textbook.

- Starting with alternative approaches. The more advanced a subject is, the fewer people will have the time and expertise to talk about it, and the more people will be tempted to expound on it without knowledge. This implies that any open discussion forum for advanced physics will be dominated by crackpots selling their personal theories.[7] When you become a researcher, you'll have to identify places where existing approaches don't work, or where previous researchers were wrong. But the path to doing so has always required learning the standard material first. Progress in physics doesn't originate from speculation in a vacuum.

[7] In the 1990s, sci.physics was an excellent discussion group full of both professors and smart amateurs. As old-timers bitterly recount, it died as crackpots came online and outnumbered the people who actually knew things. Today, sites such as Physics StackExchange and Reddit's r/Physics are only possible because of strict moderation.